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CASE STUDY ANALYSIS ON DESCRIPTIVE STATISTICS AND DATA INTERPRETATION

UNIT 3 - ASSESSMENT TOOL 1 (CO3-AT1)

DSA0404 - FUNDAMENTALS OF DATA SCIENCE

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CASE STUDY 1: EMPLOYEE SALARIES

Problem Statement

A company records the annual salaries (in ₹ lakhs) of 9 employees. The recorded salaries are as follows: 4, 5, 5, 6, 7, 8, 8, 9, and 30 lakhs. The task requires calculating various descriptive statistics to understand the salary distribution, identify outliers, and determine which measure of central tendency best represents the typical employee salary in the organization.

Dataset

Employee 1	Employee 2	Employee 3	Employee 4	Employee 5	Employee 6	Employee 7	Employee 8	Employee 9
4	5	5	6	7	8	8	9	30

Detailed Calculations

→ Mean:
$\mu = (4 + 5 + 5 + 6 + 7 + 8 + 8 + 9 + 30) / 9$
$\mu = 82 / 9$
$\mu = 9.11$ lakhs

→ Median:
Ordered data: 4, 5, 5, 6, 7, 8, 8, 9, 30
Position = $(n + 1) / 2 = (9 + 1) / 2 =$ 5th position
Median = 7 lakhs

→ Mode:
Frequency analysis: 4(1), 5(2), 6(1), 7(1), 8(2), 9(1), 30(1)
Values 5 and 8 appear twice (highest frequency)
Mode = 5 and 8 lakhs (Bimodal)

→ Range:
Range = Max(x) – Min(x)
Range = 30 – 4 = 26 lakhs

→ Variance and Standard Deviation:

Salary (x)	$(x - \mu)$	$(x - \mu)^2$
4	-5.11	26.11
5	-4.11	16.89
5	-4.11	16.89
6	-3.11	9.67
7	-2.11	4.45
8	-1.11	1.23
8	-1.11	1.23

9	-0.11	0.01
30	20.89	436.39
Sum		513.00

Variance (σ^2) = 513.00 / 9 = 57.00
Standard Deviation (σ) = $\sqrt{57.00}$ = 7.55 lakhs

→ Outlier Detection:
Using the rule: Outlier if $x > \mu + 2\sigma$ or $x < \mu - 2\sigma$
Upper limit = 9.11 + 2(7.55) = 24.21
Lower limit = 9.11 – 2(7.55) = –5.99
Value 30 > 24.21.
$\therefore$ 30 is an outlier.

→ Final Answers:

Mean	9.11 lakhs
Median	7.00 lakhs
Mode	5 and 8 lakhs (Bimodal)
Range	26 lakhs
Variance	57.00 lakhs^2
Standard Deviation	7.55 lakhs

Key Findings

→ The salary of 30 lakhs is a clear outlier, representing an executive or senior management position distinctly different from regular employees.

→ The median (7 lakhs) is the most representative measure of central tendency for typical employee salaries, as it is unaffected by the outlier.

→ The mean (9.11 lakhs) overestimates the typical salary due to the influence of the outlier, making it misleading for salary negotiation or policy decisions.

→ The range of 26 lakhs and standard deviation of 7.55 lakhs indicate substantial salary disparity within the organization.

→ The bimodal distribution suggests two distinct employee categories with different compensation levels.

CASE STUDY 2: WEBSITE RESPONSE TIME

Problem Statement

A data scientist records the response time (in milliseconds) of a website for 10 requests: 120, 125, 128, 130, 132, 135, 140, 145, 150, and 300 milliseconds. The analysis requires calculating central tendency measures, dispersion metrics, determining skewness, and evaluating why the median might be more useful than the mean in this context.

Dataset

Request 1	Request 2	Request 3	Request 4	Request 5	Request 6	Request 7	Request 8	Request 9	Request 10
120	125	128	130	132	135	140	145	150	300

Detailed Calculations

→ Mean Calculation:
$\mu = (120 + 125 + 128 + 130 + 132 + 135 + 140 + 145 + 150 + 300) / 10$
$\mu = 1385 / 10$
$\mu = 138.50$ ms

→ Median Calculation:
Ordered data: 120, 125, 128, 130, 132, 135, 140, 145, 150, 300
Position = (10 + 1) / 2 = 5.5th position
Median = (132 + 135) / 2
Median = 133.50 ms

→ Range Calculation:
Range = 300 – 120
Range = 180ms

→ Variance and Standard Deviation:

Response Time (x)	$(x - \mu)$	$(x - \mu)^2$
120	-18.50	342.25
125	-13.50	182.25
128	-10.50	110.25
130	-8.50	72.25
132	-6.50	42.25
135	-3.50	12.25
140	1.50	2.25
145	6.50	42.25
150	11.50	132.25
300	161.50	26082.25

Sum		27019.50

Variance (σ^2) = 27019.50 / 10 = 2701.95 ms^2
Standard Deviation (σ) = $\sqrt{2701.95}$ = 52.00 ms

→ Skewness Determination:
Mean (138.50) > Median (133.50)
When mean > median, the distribution is right-skewed (positively skewed).
This indicates that the outlier (300 ms) pulls the mean to the right, creating a long tail on the right side of the distribution.

→ Final Answers:

Mean Response Time	138.50 ms
Median Response Time	133.50 ms
Range	180 ms
Variance	2701.95 ms^2
Standard Deviation	52.00 ms
Skewness	Right-skewed (Positive)

Key Findings

→ The presence of one slow response (300 ms) significantly influences the mean but not the median.

→ The standard deviation of 52 ms indicates considerable variability; most requests fall around ±52 ms from the mean.

→ The range of 180 ms demonstrates a large performance gap between the fastest and slowest responses.

→ Right-skewness confirms that performance degradation occurs occasionally but is not the norm.

→ For website performance reporting to stakeholders, the median should be emphasized to accurately represent user experience.

CASE STUDY 3: STUDENT PERFORMANCE

Problem Statement

The marks of 12 students in a Data Science test are: 42, 45, 48, 50, 52, 52, 55, 58, 60, 62, 65, and 95. The analysis requires finding quartiles (Q1, Q2, Q3), calculating the interquartile range (IQR), identifying outliers using the 1.5 × IQR rule, computing mean and median, and explaining how the outlier affects the mean.

Dataset

S1	S2	S3	S4	S5	S6	S7	S8	S9	S10	S11	S12
42	45	48	50	52	52	55	58	60	62	65	95

Detailed Calculations

→ Ordered Data:
42, 45, 48, 50, 52, 52, 55, 58, 60, 62, 65, 95

→ Q1 (First Quartile) Calculation:
Position = (12 + 1) / 4 = 3.25
Q1 lies between 3rd and 4th values
Q1 = 48 + 0.25 × (50 – 48) = 48 + 0.50
Q1 = 48.50

→ Q2 (Median) Calculation:
Position = (12 + 1) / 2 = 6.5
Q2 lies between 6th and 7th values
Q2 = 52 + 0.5 × (55 – 52) = 52 + 1.50
Q2 = 53.50

→ Q3 (Third Quartile) Calculation:
Position = 3(12 + 1) / 4 = 9.75
Q3 lies between 9th and 10th values
Q3 = 60 + 0.75 × (62 – 60) = 60 + 1.50
Q4 = 61.50

→ IQR Calculation:
IQR = Q3 – Q1 = 61.50 – 48.50
IQR = 13.00

→ Outlier Detection (1.5 × IQR Rule):
Lower Bound = Q1 – 1.5 × IQR = 48.50 – 1.5 × 13.00 = 48.50 – 19.50 = 29.00
Upper Bound = Q3 + 1.5 × IQR = 61.50 + 1.5 × 13.00 = 61.50 + 19.50 = 81.00
Any value < 29 or > 81 is an outlier.
The score of 95 > 81.

∴ <u>95 is an outlier.</u>

→ Mean Calculation:
$\mu = (42 + 45 + 48 + 50 + 52 + 52 + 55 + 58 + 60 + 62 + 65 + 95) / 12$
$\mu = 684 / 12$
$\underline{\underline{\mu = 57.00}}$

→ Final Answers

Q1 (First Quartile)	48.50
Q2 (Median)	53.50
Q3 (Third Quartile)	61.50
IQR	13.00
Lower Outlier Bound	29.00
Upper Outlier Bound	81.00
Identified Outlier	95
Mean	57.00

Key Findings

→ The score of 95 is identified as a clear outlier using the 1.5 × IQR rule.
→ The median (53.50) is resistant to the outlier, making it the most reliable central tendency measure.
→ The IQR of 13.00 shows that the central 50% of students have relatively consistent performance.
→ The outlier student score of 95 increases the mean by approximately 3.45 points.
→ Q1 and Q3 effectively capture the spread of typical student performance without outlier distortion.

CASE STUDY 4: E-COMMERCE ORDERS

Problem Statement

An online store records the number of orders received over 10 days: 18, 20, 22, 22, 24, 25, 26, 28, 30, and 35 orders. The analysis requires calculating mean, median, and mode; computing variance and standard deviation; calculating the coefficient of variation; interpreting what standard deviation conveys to the company; and assessing whether the order data exhibits high or moderate variability.

Dataset

Day 1	Day 2	Day 3	Day 4	Day 5	Day 6	Day 7	Day 8	Day 9	Day 10
18	20	22	22	24	25	26	28	30	35

Detailed Calculations

→ Mean Calculation:
$\mu = (18 + 20 + 22 + 22 + 24 + 25 + 26 + 28 + 30 + 35) / 10$
$\mu = 250 / 10$
$\mu = 25.00$ orders per day

→ Median Calculation:
Ordered Data: 18, 20, 22, 22, 24, 25, 26, 28, 30, 35
Position = (10 + 1) / 2 = 5.5
Median = (24 + 25) / 2
Median = 24.50 orders per day

→ Mode Calculation:
Frequency: 22 appears twice, all others appear once
Mode = 22 orders per day

→ Variance and Standard Deviation:

Orders (x)	$(x - \mu)$	$(x - \mu)^2$
18	-7.00	49.00
20	-5.00	25.00
22	-3.00	9.00
22	-3.00	9.00
24	-1.00	1.00
25	0.00	0.00
26	1.00	1.00
28	3.00	9.00
30	5.00	25.00
35	10.00	100.00

Sum		228.00

Variance (σ^2) = 228.00 / 10 = 22.80 orders^2
Standard Deviation (σ) = $\sqrt{22.80}$ = 4.77 orders per day

→ Coefficient of Variation:
CV = (4.77 / 25.00) × 100%
CV = 19.08%

→ Final Answers:

Mean	25.00 orders/day
Median	24.50 orders/day
Mode	22 orders/day
Variance	22.80 orders^2
Standard Deviation	4.77 orders/day
Coefficient of Variation	19.08%

Key Findings

→ Average daily orders are 25, with median close at 24.50, indicating near-symmetrical distribution.

→ Standard deviation of 4.77 orders quantifies normal operational variability.

→ Coefficient of variation of 19.08% classifies this as moderate variability.

→ The highest order day (35) represents approximately +2 standard deviations from the mean.

→ The company can reasonably forecast and plan operations with this level of predictability.

CASE STUDY 5: COMPARING TWO MACHINE LEARNING MODELS

Problem Statement

Two machine learning models are evaluated based on their prediction errors across 8 test cases.
Model A errors: 2, 3, 3, 4, 4, 5, 5, 6.
Model B errors: 1, 1, 2, 4, 5, 7, 8, 9.
The analysis requires calculating mean error for both models, computing variance and standard deviation for each, determining which model makes more consistent predictions, and evaluating whether lower mean error necessarily indicates superior performance.

Dataset

Test Case	Case 1	Case 2	Case 3	Case 4	Case 5	Case 6	Case 7	Case 8
Model A Error	2	3	3	4	4	5	5	6
Model B Error	1	1	2	4	5	7	8	9

Detailed Calculations

→ Model A: Mean Error Calculation
$\mu\_A = (2 + 3 + 3 + 4 + 4 + 5 + 5 + 6) / 8$
$\mu\_A = 32 / 8$
$\underline{\mu\_A = 4.00}$

→ Model B: Mean Error Calculation
$\mu\_B = (1 + 1 + 2 + 4 + 5 + 7 + 8 + 9) / 8$
$\mu\_B = 37 / 8$
$\underline{\mu\_B = 4.63}$

→ Model A: Variance and Standard Deviation

Error (x)	**(x – μ_A)**	**(x – μ_A)²**
2	-2.00	4.00
3	-1.00	1.00
3	-1.00	1.00
4	0.00	0.00
4	0.00	0.00
5	1.00	1.00
5	1.00	1.00
6	2.00	4.00
Sum		12.00

Variance (σ^2_A) = 12.00 / 8 = 1.50
Standard Deviation (σ_A) = $\sqrt{1.50}$ = 1.22

→ Model B: Variance and Standard Deviation

Error (x)	(x – μ_B)	(x – μ_B)²
1	-3.63	13.17
1	-3.63	13.17
2	-2.63	6.91
4	-0.63	0.40
5	0.37	0.14
7	2.37	5.62
8	3.37	11.36
9	4.37	19.10
Sum		69.87

Variance (σ^2_B) = 69.87 / 8 = 8.73
Standard Deviation (σ_B) = √8.73 = 2.96

→ Model Comparison:
Model A: Mean = 4.00, SD = 1.22 (More consistent, slightly lower accuracy)
Model B: Mean = 4.63, SD = 2.96 (Higher accuracy, less consistent)
Consistency Ratio: σ_B / σ_A = 2.96 / 1.22 = 2.43
Model A is 2.43 times more consistent than Model B.

→ Final Answers:

Metric	Model A	Model B
Mean Error	4.00	4.63
Variance	1.50	8.73
Standard Deviation	1.22	2.96
Consistency Ranking	1st (More Consistent)	2nd (Less Consistent)
Better Model	Model A (Trade-off Favors Consistency)	

Key Findings

→ Model A exhibits superior consistency with a standard deviation of 1.22 versus Model B's 2.96.

→ Model A's lower mean error (4.00) reflects both accuracy and consistency advantages.

→ Model B's errors range from 1 to 9, creating unpredictability for deployment.

→ Model A's variance of 1.50 is nearly 6 times lower than Model B's 8.73.

→ In production systems, Model A's consistency makes it the superior choice despite both having moderate mean errors.